

Matteo He

ML Researcher - Mechanistic Interpretability, Language-Model Readouts, and LLM Evaluation
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Research Profile

Mechanistic interpretability researcher studying how language models learn output readouts and convert internal representations into token predictions. First author of a public arXiv preprint and a related manuscript under review, with reproducible code and pretrained artifacts; Cambridge MPhil with Distinction and production ML systems experience.

Selected Research

MPhil thesis: From Representations to Tokens: Sparse Decomposition and Training Dynamics of the Learned Readout in Language Models.

Learning to Read Out: Unembedding Dynamics [code] [artifacts]

First author; under review

- Studied when semantic information becomes natively readable during pretraining, separating hidden-state availability from output-readout expression with readout swaps, WordNet probes, and controlled tasks.
- Built parameter-trajectory crosscoders over unembedding checkpoints and released a tested pipeline, experiment manifest, ablations, guided notebooks, and pretrained artifacts.

Sparse Readout Prism [arXiv:2609.01936] [code] [models]

First-author preprint; under review

- Introduced a sparse LM-head basis that decomposes selected logits and margins into signed feature contributions with explicit residuals and a per-query fidelity gate.
- Across Qwen3.5 0.8B to 9B at selected operating points, retained **88.7-90.0% top-token agreement** on held-out states; validated reconstruction, replacement, sign, residual, and confound checks.

Automated Interpretability of GPT-2 Attention Heads

BSc dissertation, 2024

- Built a PyTorch/Transformers explainer-simulator pipeline; improved distribution-based error metrics by **25%**, reduced token errors by **57%**, and produced high-fidelity explanations for **10+** heads.

Education

University of Cambridge

MPhil in Advanced Computer Science, Distinction; AI Alignment Fellowship

Cambridge, UK

Oct 2025 - Jun 2026

University of St Andrews

BSc (Hons) Computer Science and Mathematics, First Class; GPA 18/20 (90%); Top Student Medal

St Andrews, UK

Sep 2021 - Jun 2024

Research Engineering & Industry Experience

Japanese Language Learning Platform

Founder & ML/Software Engineer

Jun 2024 - Oct 2025

Remote

- Solo-built a browser-native LLM platform serving **about 3K MAU**; eliminated server-side inference cost with WebGPU/WebLLM, cut p95 latency by **40%**, and maintained **approximately 99.9% uptime**.

Amazon, Alexa-AI (NLU)

Software Development Engineer Intern

Jun 2023 - Sep 2023

London / Cambridge, UK

- Built Python/AWS research infrastructure for **20+ Applied Scientists**, processing **millions of utterances daily**; accelerated dataset/model iteration by **15%** and reduced manual validation by **40%**.

Technical Methods

Research / ML: mechanistic interpretability, sparse representations, readout dynamics, LLM evaluation, experimental design, ablations, Python, PyTorch, Transformers, TransformerLens, JAX

Systems: GPU training, AWS S3/Lambda/DynamoDB, distributed pipelines, Docker, CI/CD, testing, WebGPU/WebLLM, TypeScript